Electrical Safety

Exam code: 0625 & 0972

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Written by: Lindsay Gilmour
Reviewed by: Caroline Carroll
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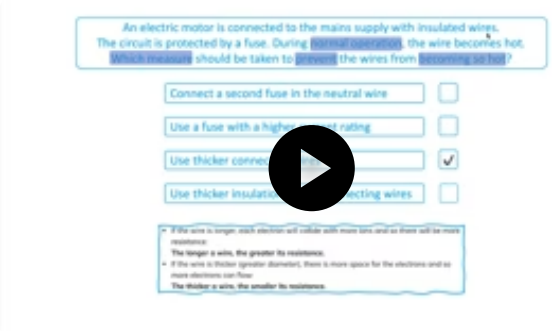

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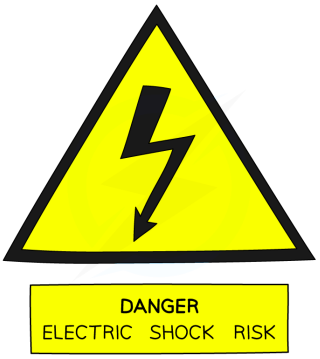
Electrical hazards

- Mains electricity can be dangerous if safety procedures are not followed
  - Voltages as low as 50 V can pose a serious risk of electrocution
- Common electrical hazards include:
  - damaged insulation
  - overheating cables
  - damp conditions
  - excess current from overloading of plugs, extension leads, single and multiple sockets when using a mains supply

Electrical danger sign



Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Damaged insulation

- If the insulation around an electrical cable is damaged, the metal part of the wire may become exposed
- If a person touches the exposed wire, they could be subjected to a lethal electric shock
- There is also a chance that current will flow between the exposed wire and any piece of metal it comes into contact with

Overheating of cables

- If an excess of current flows in a wire, this can lead to overheating
- This could cause the insulation to melt, or even cause a fire

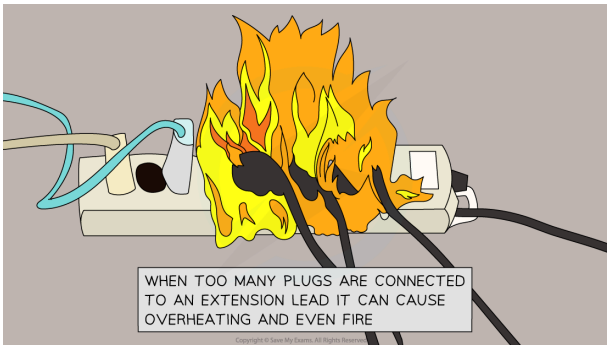
Damp conditions

- Damp conditions can be dangerous in the presence of electricity since water is an electrical conductor
- If moisture comes into contact with a live wire, this could set up
  - a short circuit within the device, which could cause a fire
  - a conductive path for current to flow through a person to the earth, which could cause electrocution

Excess current from overloading

- An excessive current may flow if too many plugs, extension leads or sockets are connected to the mains supply
- The heat created could cause the insulation to melt, or even cause a fire

Electrical fire due to excessive current



Too many appliances plugged into an extension lead can cause overheating and fire

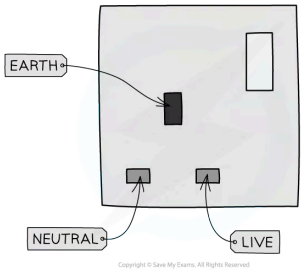
Did this video help you? Yes No

Mains Circuits

- All electrical appliances are connected to the mains supply
- A mains circuit consists of:
  - a live wire
  - a neutral wire
  - an earth wire

- The insulation covering each wire is colour-coded for easy identification:
  - Live wire – **brown**
  - Neutral wire – **blue**
  - Earth wire – **green and yellow stripes**

Live, neutral & earth wires



A mains-powered appliance in the UK contains live, neutral and earth wires

- Every country has a slightly different configuration, with many using two-pin plugs and plug sockets such as in the

- The three wires have distinct purposes

The live wire

- The purpose of the live wire is **to carry the alternating current from the mains supply to a circuit**
- It is the most dangerous of the three wires
- If it touches the appliance without the earth wire, it can cause electrocution

The neutral wire

- The purpose of the neutral wire is **to form the opposite end of the circuit to the live wire to complete the circuit**
- Because of its lower voltage, it is much less dangerous than the live wire

The earth wire

- The purpose of the earth wire is **to act as a safety wire to stop the appliance from becoming live**
- This prevents electric shocks from occurring if the appliance malfunctions or the live wire breaks off and touches the case of the plug

- In order to protect the user or the device, there are several safety features built into domestic appliances, including:
  - double insulation
  - earthing
  - fuses
  - circuit breakers

Double insulation

- The conducting part of a wire is usually made of copper or some other metal
  - If this comes into contact with a person, this poses a risk of electrocution

- For this reason, wires are covered with an insulating material, such as rubber

The conducting part of a wire is covered in an insulating material for safety

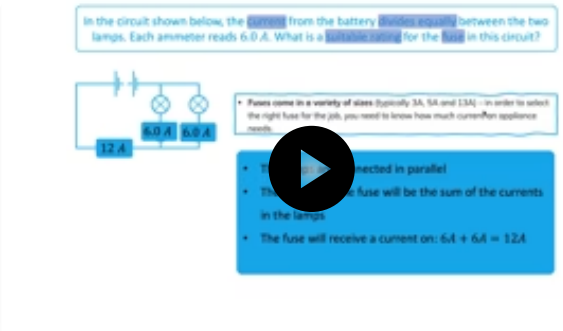
have two layers of insulation:
    - Insulation around the wires themselves
    - A non-metallic case that acts as a second layer of insulation

- Double insulated appliances do not require an earth wire or have been designed so that the earth wire cannot touch the metal casing

Earthing

- Many electrical appliances have metal cases
- This poses a potential safety hazard:
    - If a live wire (inside the appliance) came into contact with the case, the case would become electrified and anyone who touched it would risk being electrocuted

- The earth wire is an additional safety wire that can reduce this risk
- If this happens:
    - The earth wire provides a **low resistance path to the earth**
    - It causes a **surge of current in the earth wire** and hence also in the live wire
    - The high current through the fuse causes it to **melt and break**
    - **This cuts off the supply of electricity to the appliance**, making it safe



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Fuses & trip switches

- **Fuses** and **trip switches** (circuit breakers) are safety devices designed to **cut off the flow of electricity** to an appliance if the current becomes **too large** (due to a fault or a surge)

Fuses

- **Fuses** are used to protect **individual appliances**
- Fuses are located in the **plug**

cabling for a double-insulated appliance
- If the current in the wire becomes too large:
  - The wire **heats up** and **melts**
  - This causes the wire to **break**, breaking the **circuit** and **stopping** the **current**

Circuit symbol of a fuse

The circuit symbol for a fuse has a wire running through it (not to be confused with a resistor)

- Fuses have **ratings**, which signify the **maximum current** that can flow through the fuse wire without it melting
- If the current **exceeds** that value, the fuse **wire melts** and the **individual appliance** is **disconnected** from the mains supply
- Fuses come in values of **3 A**, **5 A** and **13 A**
- The correct fuse to use is the value just above the current required for the appliance

Choosing the correct fuse

- Suppose an appliance uses 3.1 amps
  - A 3 amp fuse would be **too small** - the fuse would blow as soon as the appliance was switched on
  - A 13 amp fuse would be **too large** - it would allow an extra 10 amps to pass through the appliance before it finally blew
  - A 5 amp fuse would be the **most appropriate** choice, as it is the next size up

Trip switches

- The current enters the house at the **consumer unit** (sometimes referred to as a 'fuse box')
- The consumer unit consists of a series of **trip switches** (or circuit breakers) which control the amount of current supplied to each circuit within the house
  - When the current is too high the switch 'trips' (automatically flicks to the off position)
  - This stops the current flowing in that circuit

Example of a domestic circuit

- The main **advantages** of trip switches are:
  - they provide protection from current surges or faults
  - they can be reset when the problem is fixed

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